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LAB

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Launch Amazon Linux 2 EC2 instance

we need to get source code and build so we need to install git and maven

```
yum install git java-1.8.0-openjdk maven tree -y
```

```
mvn -v
```

```
git clone https://github.com/ReyazShaik/java-project-maven-new.git
```

```
cd java-project-maven-new
```

```
ls
```

--> now we took the code and we need to build

--> if you want to see all the folders and subfolders use tree command

```
tree
```

To use Maven, maven has lifecycle called goals : each goal is a task

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MAVEN LIFECYCLE

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GOALS : is a command use to run task

Goals refer pom.xml to execute

PLUGIN: its a small software which makes our work automated

Instead of downloading tools we can download plugins

this plugins will download automatically when we run GOALS

Maven goals include compile, test, package, and install

1. mvn compile : is used to compile the code (.java[src] --> .class [target])
[in source code we have .java files, we if run compile it will create .class in target folder]

tree -- see what files are there

now, run the below command

--> mvn compile

--> tree -- it will create target folder and .class files

2. mvn test : used to test the code (.java[test]--> .class [target])

run the below command

```
mvn test
```

IT will create a test folder with .class files

```
tree
```

3. mvn package : use to create artifact

run the below command

```
mvn package
```

tree

It will show the war file, the project has frontend + backend code

4. mvn install : use to copy artifacts to .m2 (project folder --> .m2, under root/.m2 folder(maven home path))

Run the below command -- artifacts will be copied to maven home folder

mvn install

tree

5. mvn clean : to delete the target folder

Run the below command

mvn clean

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IN real time, we will not do one by one step , just use below command mvn clean package

It will generate the artifacts --> if you want to generate use below command, it will delete the existing target folder and create new

6. mvn clean package : It compile to install (all steps)

run below command

mvn clean package

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PROBLEMS WITHOUT MAVENS

--> we cant create artifacts

--> we cant create project structure

--> we cant build and deploy the apps

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Alternate tools : ANT, Gradle

Each programming language use different build tools, for example

JAVA : Maven or Gradle

PYTHON : Gradle or Maven

.NET : VS Code

C, C++ : MAKE FILE

Node js : npm (node package manager)